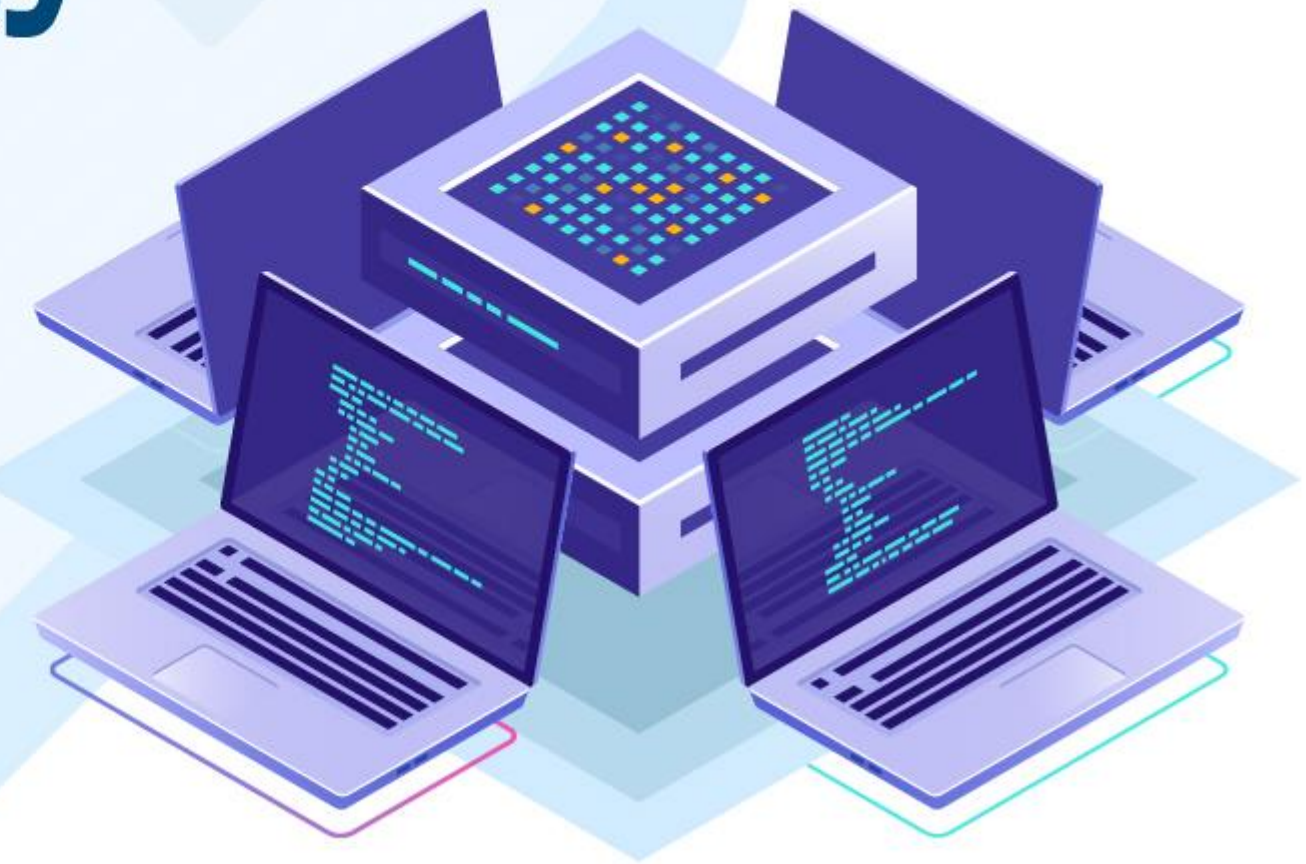
Information Technology

Year: 2026-2027

Fall Semester

- Computer Graphics
- **Dr. Yasmine Mahmoud**

Unit 2: Line drawing algorithms



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Lecture Outlines

1. Graphics Output Primitives

2. Line Drawing Algorithms
   a) Slope – Intercept Formula
   b) Parametric Form
   c) Digital Differential Analyzer (DDA)
   d) Bresenham's Line Algorithm

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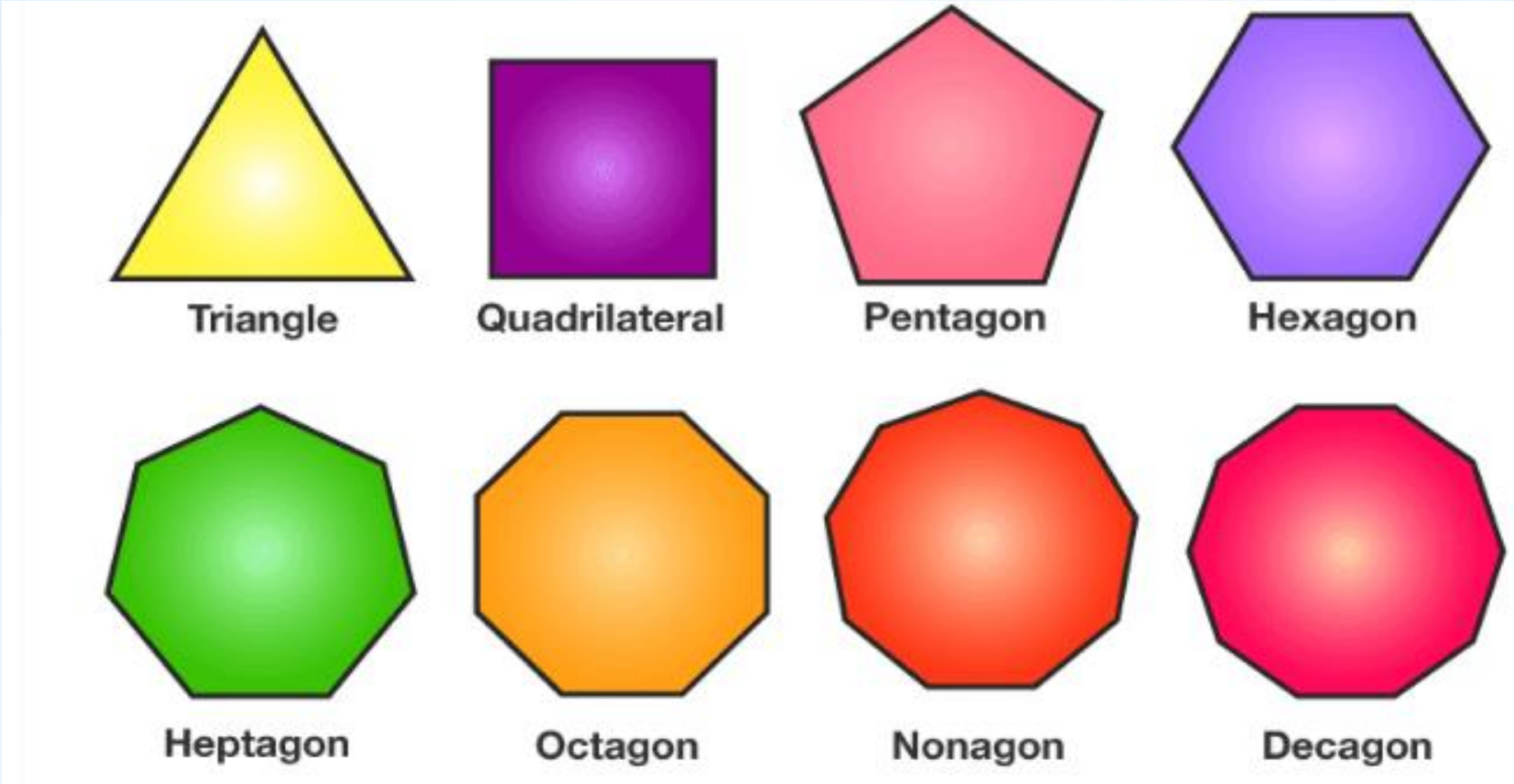
Graphics Output Primitives

Plotting Point

- Point is the fundamental element of the picture representation.
- It is nothing but the position in a plan defined as either pairs or triplets of number depending on whether the data are two or three dimensional.
- Thus, (x_1, y_1) or (x_1, y_1, z_1) would represent a point in either two, three dimensional space.
- Two points would represent a line or edge, collection of three or more points a polygon.

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Polygon Types

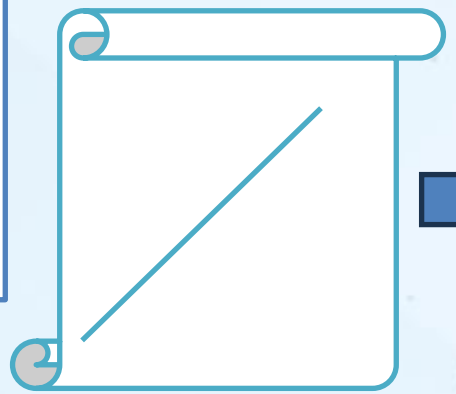


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Graphics Output Primitives

In Computer Graphics,
-The screen is divided into a group of **pixels.**

-It's not like I'm going to take a paper and a pen and draw a line on the screen.

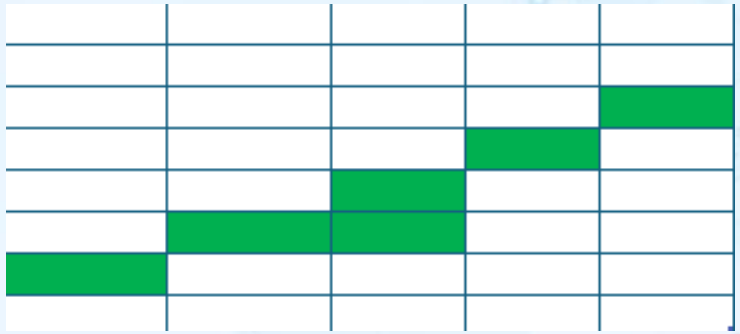


In Computer Graphics,
-When you "draw" a line, the computer doesn't draw a continuous, perfect line like on paper.

- Instead, it carefully selects which tiny squares (pixels) should be turned ON or colored to make your eyes perceive a straight line

- The **idea** is that I will **shade** (or color) some of the pixels, and finally, the shape of a straight line will appear.

- And of course, the pixels are close to each other, so the jagged (or stair-step) shape in the line doesn't appear (or is not visible).



Graphics Output Primitives

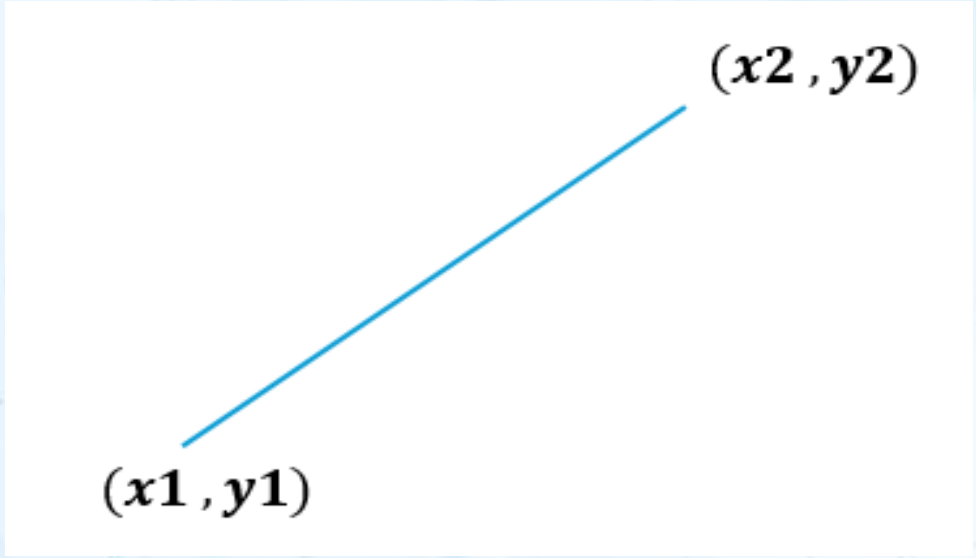
Generally,

- I have two end points and I want to draw a line between them.

- What does that mean?

- It means I will use an algorithm to generate the set of points that are between the two end points.

- Finally, the shape of the straight line will appear.



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Graphics Output Primitives

The general equation of any straight line is:

$$y = mx + c$$

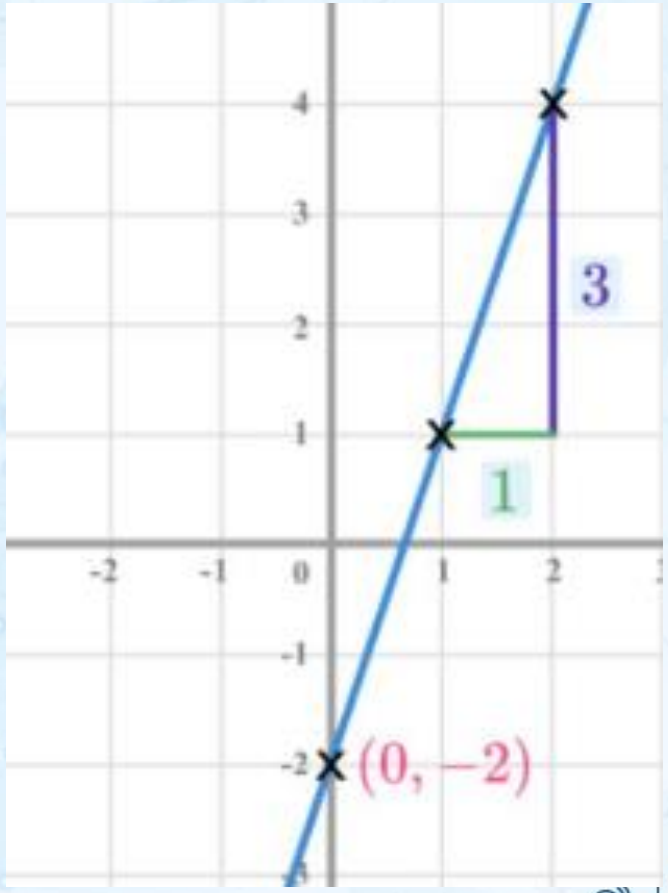
c is the **y-intercept** (where the line crosses the y-axis)

m is the **gradient** (steepness) of the line

Example The graph of the line $y = 3x - 2$

The gradient is 3

The y-intercept is -2, the coordinate $(0, -2)$

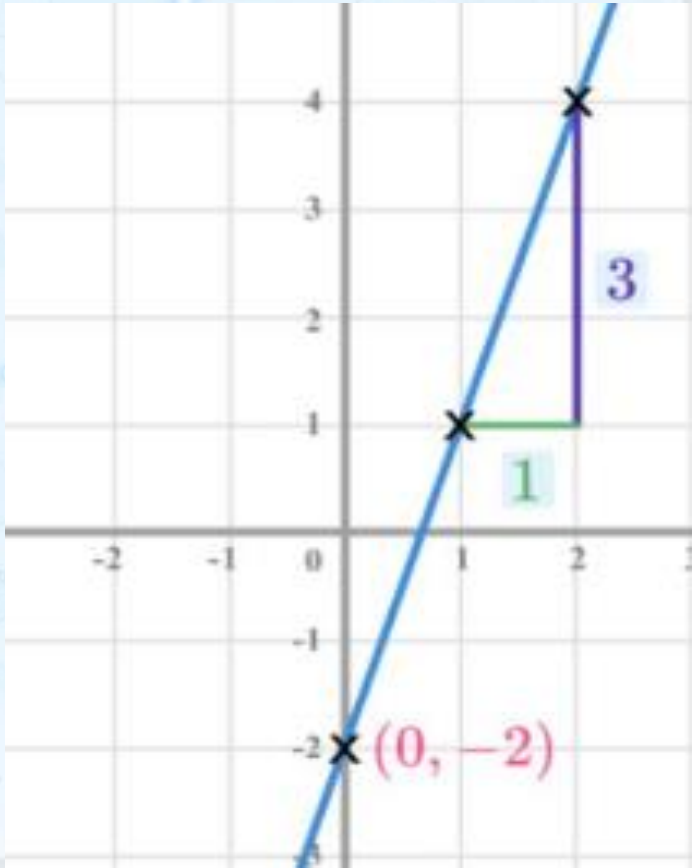


Graphics Output Primitives

- If the **two** points used to specify a line are (x_1, y_1) and (x_2, y_2), then an equation for the line is given as:

- $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1} = \mathrm{m}$

- $y = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1) + y_1$



Graphics Output Primitives

- **Y= $mx + b$** , This equation is called the slope- intercept form of the line. Where, $m = \frac{y_2 - y_1}{x_2 - x_1}$ and $b = y_1 - mx_1$

  - The slope m is the change in height ($y_2 - y_1$) divided by the change in the width ($x_2 - x_1$) for two points on the line.

  - The intercept b is the height at which the line crosses the y-axis.

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Graphics Output Primitives

Example :

- Write an equation in slope-intercept form of the line that passes through the points (2 , 2) and (3 , 4). **m** represents the slope.

- **4 is y_2 and 2 is y_1 3 is x_2 and 2 is x_1**

- $m = \frac{y_2 - y_1}{x_2 - x_1}$, $m = \frac{4-2}{3-2} = \frac{2}{1}$, $m = 2$

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Graphics Output Primitives

Example :

- Use the slope-intercept form of a line and the values we know to solve for **b**, which is the y-intercept.

$Y = mx + b \quad , \quad Y = 2x + b$

$4 = 2(3) + b \quad , \quad 4 = 6 + b$

- $b = 4 - 6 \quad , \quad b = -2$
- b is equal to -2 , y-intercept
- Plug the values for slope and y-intercept into the slope-intercept form to find the equation of our line
- $Y = mx + b$
- $Y = \mathbf{2x - 2}$

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Graphics Output Primitives

Write the slope-intercept form of the equation of the line that passes through the two points.

1. (2, 3), (6, 11)

$y = 2x - 1$

2. (1, -7), (3, -15)

$y = -4x - 3$

Write an equation of the line in point-slope form that passes through the point and has the given slope. Then rewrite the equation in slope-intercept form.

3. (-1, 1), $m = \frac{2}{3}$

$y = \frac{2}{3}x + \frac{5}{3}$

4. (6, -3), $m = -\frac{1}{2}$

$y = -\frac{1}{2}x$

Write the equation in standard form of the line that passes through the two points.

5. (5, 8), (3, 2)

$3x - y = 7$

6. (-4, -5), (-2, 5)

$5x - y = -15$

Graphics Output Primitives

3. $(-1, 1)$, $m = \frac{2}{3}$

$$y = \frac{2}{3}x + \frac{5}{3}$$

$x_1\ y_1$

$b = y_1 - m x_1$

$= 1 - \left(\frac{2}{3} * -1\right)$

$= 1 + \frac{2}{3}$

$= \frac{3+2}{3} = \frac{5}{3}$

4. $(6, -3)$, $m = -\frac{1}{2}$

$$y = -\frac{1}{2}x$$

$x_1\ y_1$

$b = y_1 - m x_1$

$= (-3) + \left(\frac{1}{2} * 6\right)$

$= -3 + \frac{6}{2}$

$= -3 + 3 = \text{Zero}$

$b = \text{Zero}$

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Graphics Output Primitives

Write the equation in standard form of the line that passes through the two points.

5. (5, 8), (3, 2)

$3x - y = 7$

$m = \frac{2-8}{3-5} = \frac{-6}{-2}$

$m = 3$

$b = y_1 - mx_1$

$= 8 - (3 * 5)$

$= -7$

$y = 3x - 7$

6. (-4, -5), (-2, 5)

$5x - y = -15$

$m = \frac{5-(-5)}{-2-(-4)} = \frac{10}{2} = 5$

$b = y_1 - mx_1$

$= -5 - (5 * -4)$

$b = -5 + 20 = 15$

$y = 5x + 15$

$5x - y = -15$

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Graphics Output Primitives

- It can determine whether two lines are crossing or parallel.
- When two lines cross, they share some point in common.
- That point satisfies equations for the two lines.
- How to obtain this point?
- Consider two lines having slops $\boldsymbol{m1}$ and $\boldsymbol{m2}$ and y-intercepts $\boldsymbol{b1}$ and $\boldsymbol{b2}$ respectively.
- Then we can write the equations for these two lines as,
  - **Line 1:** $y = m_1 x + b_1$
  - **Line 2:** $y = m_2 x + b_2$



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Graphics Output Primitives

- If point (x_p, y_p) is shared by both lines, then it should satisfy the last two equations.
- And, $\boldsymbol{y_p = m_1 x_p + b_1}$, $\boldsymbol{y_p = m_2 x_p + b_2}$

- Equating the above equations gives,
- $\boldsymbol{m_1 x_p + b_1 = m_2 x_p + b_2}$
- $\boldsymbol{m_1 x_p - m_2 x_p = b_2 - b_1}$
- $\boldsymbol{x_p(m_1 - m_2) = b_2 - b_1}$

- $\boldsymbol{x_p = \frac{b_2 - b_1}{m_1 - m_2}}$

Graphics Output Primitives

$$x_p = \frac{b_2 - b_1}{m_1 - m_2}$$

- Substituting this into equation for line 1 or for line 2 gives,
- $y_p = m_1 \left[\frac{b_2 - b_1}{m_1 - m_2}\right] + b_1$

- $y_p = \frac{m_1 (b_2 - b_1) + b_1 (m_1 - m_2)}{m_1 - m_2}$
- $y_p = \frac{m_1 b_2 - m_1 b_1 + b_1 m_1 - b_1 m_2}{m_1 - m_2}$
- $y_p = \frac{m_1 b_2 - b_1 m_2}{m_1 - m_2}$
- Therefore, the point $\left[\frac{b_2 - b_1}{m_1 - m_2}, \frac{m_1 b_2 - m_2 b_1}{m_1 - m_2}\right]$ is the point of intersection.

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If point (xp,yp) is shared by both lines, then it should satisfy the last two equations (point of intersection).

How to get the intersection point (Mathematical Proof)

$$yp = m1\,xp + b1$$

$$yp = m2\,xp + b2$$

So, $$m1\,xp + b1 = m2\,xp + b2$$

$$m1\,xp - m2\,xp = b2 - b1$$

$$xp = (b2 - b1) / (m1 - m2)$$

therefore, $$yp = m1\,xp + b1$$

then, $$yp = m1\,(\frac{(b2-b1)}{m1-m2}) + b1$$

$$yp = \frac{m1\,b2 - m1\,b1}{m1 - m2} + b1\,\frac{m1-m2}{m1-m2}$$

$$yp = \frac{m1\,b2 - m1\,b1 + m1\,b1 - m2\,b1}{m1 - m2}$$

$$yp = \frac{m1b2 - m2b1}{m1 - m2}$$

the intersection point $(xp\,,yp)$ is :

$$[\frac{(b2-b1)}{m1-m2}\,,\frac{m1\,b2-m2\,b1}{m1-m2}]$$

If point (xp,yp) is shared by both lines, then it should satisfy the last two equations (point of intersection).

How to get the intersection point (Mathematical Proof)

$$yp = m1\,xp + b1$$

$$yp = m2\,xp + b2$$

So, $$m1\,xp + b1 = m2\,xp + b2$$

$$m1\,xp - m2\,xp = b2 - b1$$

$$xp = (b2 - b1) / (m1 - m2)$$

therefore,

$$yp = m2\,xp + b2$$

then, $$yp = m2\,(\frac{(b2-b1)}{m1-m2}) + b2$$

$$yp = \frac{m2\,b2 - m2\,b1}{m1 - m2} + b2\,\frac{m1-m2}{m1-m2}$$

$$yp = \frac{m2\,b2 - m2\,b1 + m1\,b2 - m2\,b2}{m1 - m2}$$

$$yp = \frac{m1b2 - m2b1}{m1 - m2}$$

the intersection point $(xp\,,yp)$ is :

$$[\frac{(b2-b1)}{m1-m2}\,,\frac{m1\,b2-m2\,b1}{m1-m2}]$$

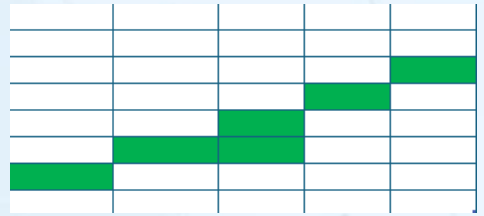
Graphics Output Primitives



- If the two lines are parallel, they have the same slope.
- In the expression for the point of intersection, $\boldsymbol{m1 = m2}$.
- So, the denominator becomes zero.
- So, the expression results in a divide by zero.
- This means, there is no point of intersection for the parallel lines.

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Line Drawing Algorithms



- The process of 'turning on' the pixels for a line segment is called vector generation or line generation, and the algorithms for them are known as vector generation algorithms or line drawing algorithms.

- These requirements specify the desired characteristics of line:
  - The line should appear as a straight line and it should start and end accurately.
  - The line should be displayed with constant brightness along its length **independent** of its length and orientation.
  - The line should be drawn rapidly.

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Line Drawing Algorithms

* Slope - Intercept Formula (1st Technique)

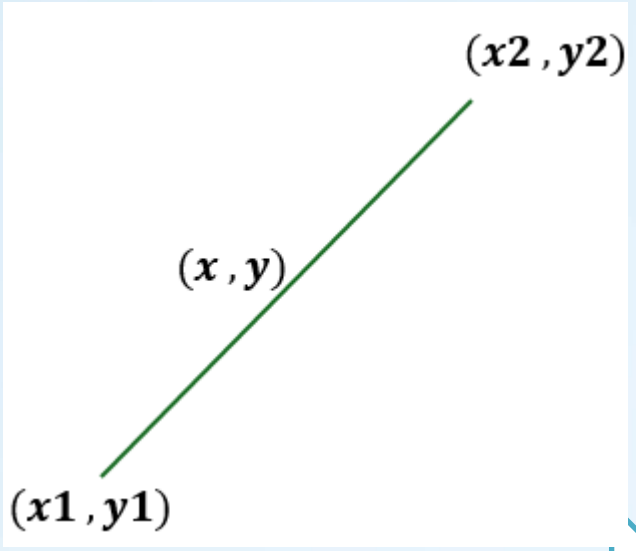
- It is a traditional mathematical method that has been used a lot in the field of mathematics to draw a straight line between any two end points.

- The whole idea is that the **slope** (or **gradient**) of a straight line is **always constant** (or **fixed**).

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Line Drawing Al

Explanation



For example,

- I have a line that **started** at **Point (x1 , y1)** and **ended** at **Point (x2 , y2)**, and there is a point **(x , y)** on the line.

- Therefore, we can **determine** the **slope** of that straight line = $\frac{\Delta y}{\Delta x}$ (meaning the **difference** between the **Y-coordinates** (Δy) and the **X-coordinates** (Δx).

- Slope = **m** = $\frac{y2 - y1}{x2 - x1}$, **at the same time** $m = \frac{y - y1}{x - x1}$

- **Then,** $\frac{y2 - y1}{x2 - x1} = \frac{y - y1}{x - x1}$

- To get **Y**, we will **move** the **'Y'** to the **other side** of the previous equation.

$$y = y1 + \left(\frac{y2 - y1}{x2 - x1}\right) x - \left(\frac{y2 - y1}{x2 - x1}\right) x1$$

Since the **equation** has **constants**, I will use the equation in **this form.**

$$y = M x + B$$

$$M = \frac{y - y1}{x - x1}$$

$$B = y1 - \left(\frac{y - y1}{x - x1}\right) x1$$

Y= $mx + b$, This equation is called the **slop intercept form** of the line. The intercept *b* is the height at which the line crosses the y-axis.

Line Drawing Algorithms

Example

- Draw a straight line from point (10 , 12) to point (40 , 20)

- To draw a line, first calculate M & B.

1. $\mathbf{M = \frac{y - y1}{x - x1} = \frac{20 - 12}{40 - 10} = \frac{8}{30}}$

2. B = y1 - ($\frac{y - y1}{x - x1}$) x1 **Therefore,** $\frac{y2 - y1}{x2 - x1} = \frac{y - y1}{x - x1}$

Then, B = 12 - ((8/30) * 10) = 28/3

3. y = M x + B

y = ((8/30) * X)+ (28/3)

4. I will start drawing **starting** from the first point (10 , 12) up to the last point (40 , 20) by substituting the X-values in the equation **y = ((8/30) * X)+ (28/3)** starting from X =10 (as the) first point (10 , 12) .

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Line Drawing Algorithms

X value	Y value
10	12
11	12.266
12	
40	20



The problem with this method

-is the **decimal numbers** (or **floating-point numbers**) and their **representation** (or **calculation**), which is **very expensive** (or **resource-intensive**).

-All of this makes the work in this field of computer graphics **very slow**.

-This method is **not suitable** for the **graphics field** because, in this field, we need to see things **instantaneously** (or **in real-time**) on the screen, we don't want to see a line being drawn **point by point slowly.**

Line Drawing Algorithms

* Parametric Form (2nd Technique)

- t is a varaiable between (zero and one)

- so that, If **t = 0**

then, **x = x1 + 0 (x2 - x1)**

so that, **x = x1**

&

y = y1 + 0 (y2 - y1)

so, y = y1

Note that, if x = x1 & y = y1 , so that when t = 0, after substitution we will get the first point (**start point**)

General Form:

x = x1 + t (x2 - x1)

&

y = y1 + t (y2 - y1)

- t is a varaiable between (zero and one)

- so that, If **t = 1**

then, **x = x1 + 1 (x2 - x1)**

so that, **x = x2**

&

y = y1 + 1 (y2 - y1)

so, y = y2

Note that, if x = x2 & y = y2 , so that when t = 1, after substitution we will get the last point (**end point**)

- If we put **'t'** (or a **parameter**) with **any value between one and zero**, **0 <= t <= 1** ,I will get the **points** of the **straight line** between the **first point** and the **last point** of the line.

- Determine **how many points** you want to draw a line with.
- ex. if **10 points**, start with t = 0 , t = 0.1 , t = 0.2 , ... , t=1
- ex. if **100 points** (to show accurate enough), start with t = 0 , t = 0.01 , t = 0.02 , ... , t=1

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Line Drawing Algorithms

* Parametric Form (2nd Technique)

Example

- Draw a line from point (5 , 10) to point (25 , 18)

Solution:

First, write both equations,

$$x = x1 + t\,(x2 - x1)$$

$$y = y + t\,(y2 - y1)$$

Then substitute on both:

$$x = 5 + t\,(25 - 5)$$

$$x = 5 + 20\,t$$

$$y = 10 + t\,(18 - 10)$$

$$y = 10 + 8\,t$$

Then, draw the table

t	x	y	Point (x , y)
0	5	10	(5 , 10)
0.01	5.2	10	(5.2 , 10)
0.02	5.4	10	
0.03	5.6	10	
1	25	18	(25 , 18)

What is the problem about this method??

- Of course, this method (**The Parametric Form**) is better than the one before it (The direct equation y=mx+b).
- but the problem is that we **still** use **decimal numbers** (or **floating-point numbers**) and **multiplication operations**, which are also **costly**.

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QUESTIONS?

